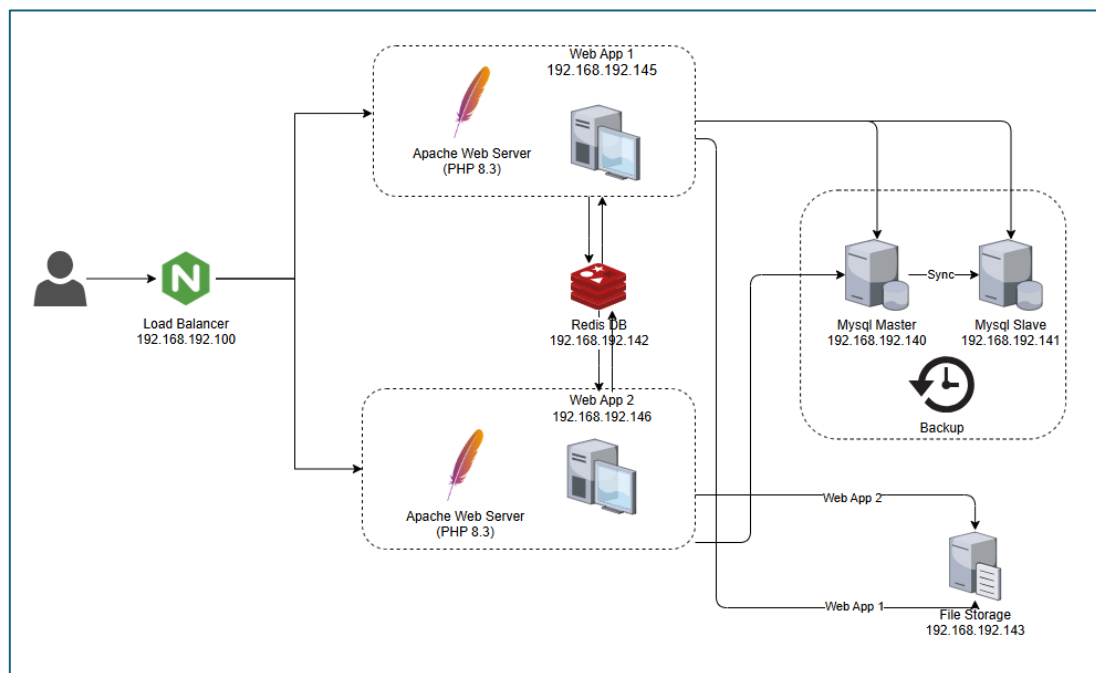


Full Architecture System

1. Support High Availability / redundancy
2. Uptime 99.9%
3. Ensure network traffic

System Architecture



Network Setup:

- 1) `sudo nano /etc/netplan/50-cloud-init.yaml`
- 2) `sudo netplan apply`

```
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.192.xxx/24 ## IP Static (.100> frontdoor / .140> backdoor)
      routes:
        - to: default
          via: 192.168.192.2
      nameservers:
```

addresses: [8.8.8.8, 8.8.4.4]

VM Name	Purpose	IP Setup
Nginx	Load Balancer	192.168.192.100
Web-App-1	Apache Node	192.168.192.145
Web-App-2	Apache Node	192.168.192.146
MySQL-Master	Database	192.168.192.140
MySQL-Slave	Database	192.168.192.141
Redis-DB	Caching	192.168.192.142
Storage-NFS	Files	192.168.192.143

Purposes

Component	IP Address	Service	Primary Role
Load Balancer	192.168.192.100	Nginx	Entry point; distributes traffic via upstream.
Web App 1	192.168.192.145	Apache / PHP	Active processing node (Primary).
Web App 2	192.168.192.146	Apache / PHP	Active processing node (Redundant).
Redis	192.168.192.142	Redis Server	Centralized session storage for user persistence.
MySQL Master	192.168.192.140	MySQL 8.0	Primary database for all Write operations.
MySQL Slave	192.168.192.141	MySQL 8.0	Read-only replica for data redundancy.
NFS Storage	192.168.192.143	NFS Kernel	Shared file system for static assets and uploads.

Configuration

MySQL Master & Slave

Master (.140)

1. `sudo nano/etc/mysql/mysql.conf.d/mysqld.cnf`

```
server-id = 1
log_bin = /var/log/mysql/mysql-bin.log
bind-address = 0.0.0.0
```

```
CREATE USER IF NOT EXISTS 'kyrol'@'192.168.192.%' IDENTIFIED
BY 'your_password'; GRANT ALL PRIVILEGES ON cluster_test.* TO
'kyrol'@'192.168.192.%'; FLUSH PRIVILEGES;
```

Slave (.141)

1. `sudo nano/etc/mysql/mysql.conf.d/mysqld.cnf`

```
server-id = 2
relay-log = /var/log/mysql/mysql-relay-bin.log
bind-address = 0.0.0.0
```

NFS Storage

1. `/etc/exports` (shares the folder with specific web app subnet)
2. `/mnt/shared_uploads 192.168.192.0/24(rw,sync,no_subtree_check)`
3. On web app 1 & 2 = `sudo mount 192.168.192.143:/mnt/shared_uploads /var/www/html/uploads`

Nginx Load Balancer

1. Create new file on `/etc/nginx/sites-available/` on this case **web_lb**
2. Config:

```
upstream my_web_cluster {
    # Distributes traffic between your two Apache nodes
    least_conn;
    server 192.168.192.145; # Web App 1
    server 192.168.192.146; # Web App 2
}

server {
    listen 80;
    server_name _;
```

```

    location / {
        proxy_pass http://my_web_cluster;

        # Passes the real visitor's info to the back-end
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For
$proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}

```

Redis Session Configuration

1. `sudo nano /etc/redis/redis.conf`

```

# Listen on all network interfaces
bind 0.0.0.0

# Disable protected mode to allow external connections
protected-mode no

```

2. Inside Web App 1 & 2

`sudo nano /etc/php/8.3/apache2/php.ini`

```

session.save_handler = redis
session.save_path = "tcp://192.168.192.142:6379" #Redis IP

```

Web App

1. `sudo hostnamectl set-hostname web-app-2`
2. `sudo nano /etc/hosts` # **Config for Web App 2 ONLY**

Install PHP & Apache

1. `sudo apt update`
2. `sudo apt install apache2 php libapache2-mod-php php-mysql php-redis php-gd php-curl php-xml php-mbstring -y`

Configure Header

1. `sudo nano /etc/apache2/conf-available/remoteip.conf`

```

RemoteIPHeader X-Forwarded-For
RemoteIPInternalProxy 192.168.192.144 # Nginx LB

```

2. **Enable remoteIP module** : `sudo a2enmod remoteip`

3. **Enable your specific configuration :** `sudo a2enconf remoteip`
4. **Test and Restart :** `sudo apache2ctl configtest && sudo systemctl restart apache2`

Configure Redis - Web App

1. `sudo nano /etc/php/8.3/apache2/php.ini`

```
session.save_handler = redis  
session.save_path = "tcp://192.168.192.142:6379" # Redis IP
```

2. `sudo systemctl restart apache2`

Configure NFS Mount

1. Install NFS Client : `sudo apt install nfs-common -y`
2. Create mount point : `sudo mkdir -p /var/www/html/uploads`
3. Configure auto-mount of boot : `sudo nano /etc/fstab`

```
192.168.192.143:/mnt/shared_uploads /var/www/html nfs defaults  
0 0
```

4. Mount everything : `sudo mount -a`

Configure Permission

1. Set Apache Ownership : `sudo chown -R www-data:www-data /var/www/html/uploads`
2. Set folder permissions : `sudo chmod -R 755 /var/www/html/uploads`

Setting	Web App 1	Web App 2
Hostname	web-app-1	web-app-2
IP Address	192.168.192.145	192.168.192.146
PHP Version	8.3	8.3
Session Store	Redis (.142)	Redis (.142)
Data Source	MySQL Master (.140)	MySQL Master (.140)
Uploads	NFS Storage (.143)	NFS Storage (.143)

Documentation

MySQL

```
mysql> SHOW SLAVE STATUS\G
Replicate_Ignore_DB:
Replicate_Do_Table:
Replicate_Ignore_Table:
Replicate_Wild_Do_Table:
Replicate_Wild_Ignore_Table:
Last_Errno: 0
Last_Error:
Skip_Counter: 0
Exec_Master_Log_Pos: 157
Relay_Log_Space: 759
Until_Condition: None
Until_Log_File:
Until_Log_Pos: 0
Master_SSL_Allowed: No
Master_SSL_CA_File:
Master_SSL_CA_Path:
Master_SSL_Cert:
Master_SSL_Cipher:
Master_SSL_Key:
Seconds_Behind_Master: 0
Master_SSL_Verify_Server_Cert: No
Last_IO_Errno: 0
Last_IO_Error:
Last_SQL_Errno: 0
Last_SQL_Error:
Replicate_Ignore_Server_Ids:
Master_Server_Id: 1
Master_UUID: 6149cb46-3176-11f1-bc6f-000c290df1ac
Master_Info_File: mysql.slave_master_info
SQL_Delay: 0
SQL_Remaining_Delay: NULL
Slave_SQL_Running_State: Replica has read all relay log; waiting for more updates
Master_Retry_Count: 86400
Master_Bind:
Last_IO_Error_Timestamp:
Last_SQL_Error_Timestamp:
Master_SSL_Crl:
Master_SSL_Crlpath:
Retrieved_Gtid_Set:
Executed_Gtid_Set:
Auto_Position: 0
Replicate_Rewrite_DB:
Channel_Name:
Master_TLS_Version:
Master_public_key_path:
Get_master_public_key: 0
Network_Namespace:
1 row in set, 1 warning (0.01 sec)

mysql> _
```

Nginx LB

```
kyrol@kyrol:~$ sudo nginx -t
[sudo] password for kyrol:
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
kyrol@kyrol:~$
```

Redis

```
kyrol@mysql-master:~$ ss -tln | grep 6379
tcp    LISTEN 0      511      0.0.0.0:6379      0.0.0.0:*
kyrol@mysql-master:~$
```

NFS (Web app 1 & Web app 2)

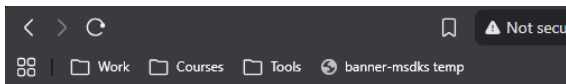
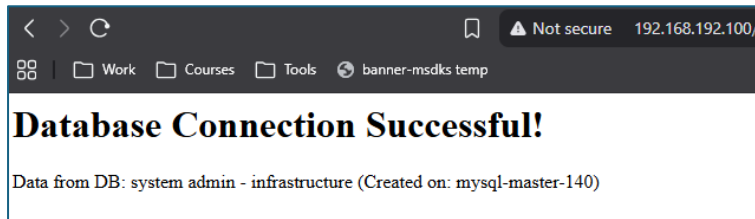
```
Filesystem            Size      Used Avail Use% Mounted on
tmpfs                  193M      1.3M   191M    1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G      4.6G      4.7G   50% /
tmpfs                  961M        0    961M    0% /dev/shm
tmpfs                  5.0M        0    5.0M    0% /run/lock
/dev/sda2              1.7G     104M     1.5G    7% /boot
192.168.192.143:/mnt/shared_uploads 9.8G      5.3G      4.1G   57% /var/www/html
tmpfs                  193M      12K    193M    1% /run/user/1000
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

```
kyrol@kyrol:~$ df -h
Filesystem            Size      Used Avail Use% Mounted on
tmpfs                  193M      1.3M   191M    1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G      4.6G      4.8G   50% /
tmpfs                  961M        0    961M    0% /dev/shm
tmpfs                  5.0M        0    5.0M    0% /run/lock
/dev/sda2              1.7G     104M     1.5G    7% /boot
192.168.192.143:/mnt/shared_uploads 9.8G      5.3G      4.1G   57% /var/www/html/uploads
tmpfs                  193M      12K    193M    1% /run/user/1000
kyrol@kyrol:~$
```


Web App DB Connection

Condition 1: Both Web App running

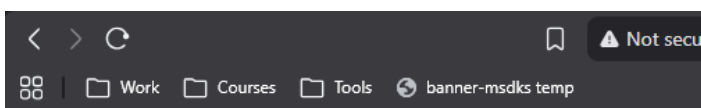
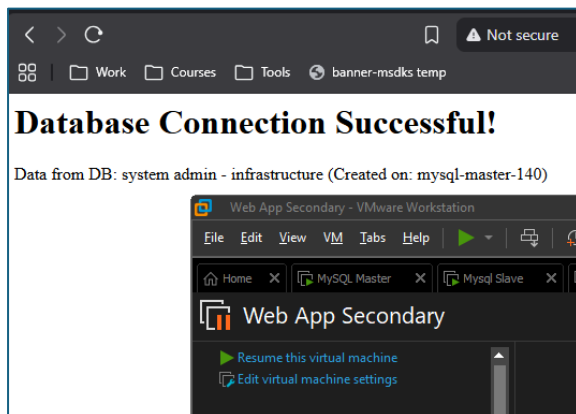


Connected to Redis successfully!
Ping Response: 1

Redis Server Info:

Redis Version: 7.0.15
Connected Clients: 1
Used Memory: 943.45K

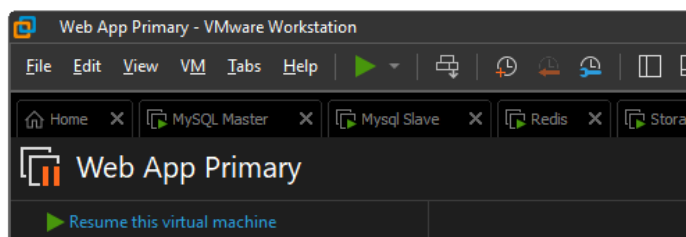
Condition 2: Primary Web App running



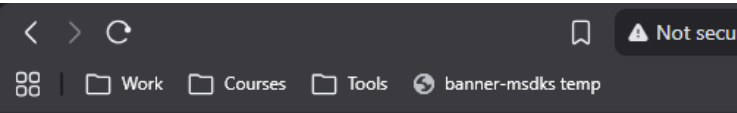
Connected to Redis successfully!
Ping Response: 1

Redis Server Info:

Redis Version: 7.0.15
Connected Clients: 1
Used Memory: 943.45K

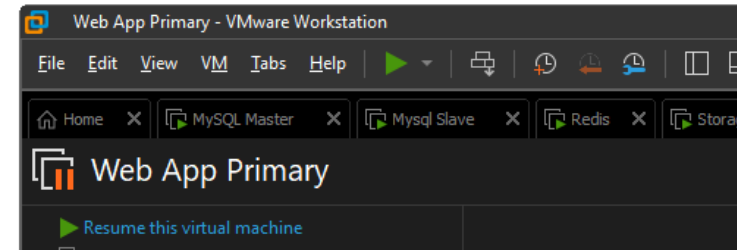


Condition 3: Web App 2 Running



Redis Server Info:

Redis Version: 7.0.15
Connected Clients: 1
Used Memory: 943.45K



Database Connection Successful!

Data from DB: system admin - infrastructure (Created on: mysql-master-140)

